



SHORT SanDisk Corporation (NASDAQ: SNDK)

Current Price: **\$1,764** (09/09/2026)

Target Price: **\$927** (-47% downside)

For Varsity Pitch Competition 2026

The market has capitalised a five-year contract as a perpetuity

Oxford Alpha Fund · Varsity Pitch 2026
Prices and consensus as of 9 Sep 2026

Recommendation and thesis

Price (9 Sep 2026) · market cap · EV	\$1,764 · \$258bn · \$254bn
12-month base target · scenario range	\$927 (-47%) · \$627 - \$2,264
Probability-weighted short return · reward/risk	+28% · 4.3 : 1
Intrinsic value (Monte Carlo median) · price percentile	\$663 · 97th
P/E FY27E · FY28E · price/book	8.2× · 6.7× · 16.7×
Short interest · borrow · β vs SMH	5.3% · 0.28% · 1.73

Pay the Street's boom in full, pay the \$93.9bn contract floor in full through FY31, and today's price still needs **\$26.2bn of profit a year, forever**, from a business whose best pre-2026 year earned **\$1.6bn**. The contracts have an expiry date; the price does not.

1 The floors are far below consensus

\$93.9bn over ~4.5 years is \$20.9bn/yr for ~2/3 of bits. Floor pricing on every bit is \$31.1bn/yr, **-46%** vs FY28 consensus of \$57.8bn. Only \$2.7bn of the \$16.5bn of "guarantees" is on SanDisk's balance sheet.

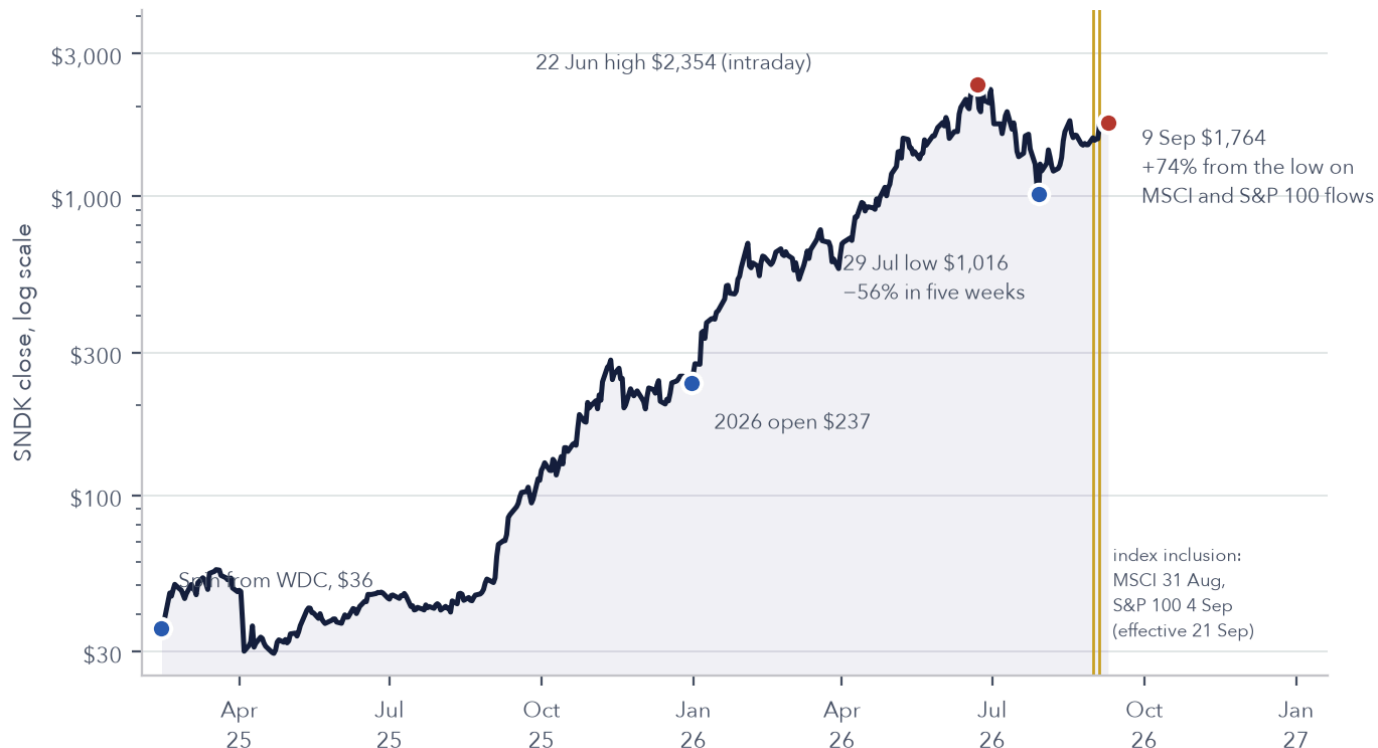
2 The earnings are price, and price has stopped accelerating

Consensus FY27 is +142% revenue on +15% bits. Contract prices went +55-60% → +70-75% → +10-15% QoQ; August +1.4%; spot wafers now falling. Every dated primary forecast puts the supply crossover in 2H27.

3 Even the bulls' normalised numbers do not reach the price

Morgan Stanley's through-cycle EPS needs a 23× multiple to justify \$1,764. Across 33 documented NAND company-years the median operating margin is 9%; management's model assumes ~75% indefinitely. Kioxia, co-owner of the same fabs, trades at 4.4× earnings vs SanDisk's 8.2×

\$36 → \$2,354 → \$1,016 → \$1,764 in nineteen months



Why the mispricing exists

A true four-year story ("contracted, non-cyclical") priced as permanence; estimates marked to spot; index and pod ownership after a 30× market-cap increase; an 18-month-old listing with no memory of losing \$2.1bn in a bad year.

Why it is uncrowded

Short interest 5.3% of float (FINRA 14 Aug), ~1 day to cover, borrow 0.28% GC; 20 of 24 analysts at Buy, mean target \$2,125; 128 hedge funds hold \$25.6bn. Druckenmiller and Tepper exited in Q2.

Why now

Micron prints 30 Sep, TrendForce's first 4Q26 NAND number early Oct, hyperscaler Q3 late Oct, SanDisk FQ1 on 6 Nov. Entry after the 21 Sep S&P 100 inclusion.

A fab-less NAND pure-play whose income statement is a price, with \$93.9bn of contracted floors through FY31

The business in brief

- **NAND flash only.** Spun out of Western Digital on 13 Feb 2025 at \$36. No DRAM, no HBM. Products: enterprise SSD, client SSD, consumer cards and drives, embedded mobile. Datacenter is ~38% of bits and the growth engine; Consumer fell 32% QoQ in FQ4.
- **It does not own its fabs.** Every wafer comes from Flash Ventures, the 49.9% JV with Kioxia (Yokkaichi, Kitakami). SanDisk pays **half of the JV's fixed costs regardless of output**; \$6.6bn of JV commitments FY27-31; \$923m lease guarantees. Capex on its own books is ~\$0.2bn a year.
- **FY26 (Jun):** revenue \$20.2bn (+175%), gross margin 71.5%, operating income \$12.5bn, net income \$11.4bn, FCF \$11.5bn; cash \$4.8bn, debt \$0.2bn; \$4.5bn bought back at ~24× trailing; \$15.5bn authorisation remains.
- **FQ1-FY27 guide:** revenue \$10.5-11.2bn, gross margin **83-85%**, opex \$520-540m per quarter, non-GAAP tax 15%.

FY (Jun), \$bn	FY23	FY24	FY25	FY26	FY27E	FY28E
Revenue	6.1	6.7	7.4	20.2	49.0	57.8
Gross margin	7%	16%	30%	71%	~83%	~80%
Operating income	-1.3	-0.4	0.5	12.5	-	-
Net income	-2.1	-0.7	-1.6	11.4	\$31.3bn	\$38.8bn
EPS · P/E at \$1,764	-14.78	-4.63	-11.32	73.7 · 24×	\$214 · 8.2×	\$265 · 6.7×

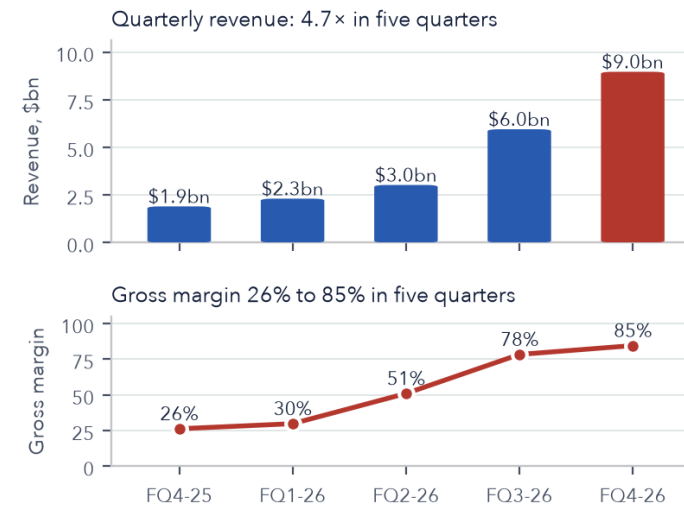
Three losses in three years, then the best year any NAND company has printed. FY27-28 are Yahoo/LSEG consensus (19-20 analysts); FY28 revenue estimates span \$42.6bn to \$83.4bn.

What the FQ1 guide annualises to

FQ1-FY27 guide, midpoint	Quarter	× 4
Revenue \$10.5-11.2bn	\$10.85bn	\$43.4bn
Gross profit at 84% GM	\$9.1bn	\$36.5bn
Operating income after \$530m opex	\$8.6bn	\$34.4bn
Net income at 15% tax · EPS	\$7.3bn · \$50	\$29.2bn · \$200

Consensus FY27 (\$49.0bn revenue, \$214 EPS) needs pricing to keep rising through FY27, not just hold (slide 7).

Five quarters that made the multiple



Capital allocation at the top of the cycle

FY26 cash	\$bn	Note
Free cash flow	11.5	from -0.1 in FY25
Buyback executed (FQ4)	4.5	at ~24× trailing earnings
Authorisation remaining	15.5	≈ 6% of shares at \$1,700
JV commitments FY27-31	6.6	unconditional
Share of ¥5tn Japan plan	~15	Fab 3 output FY2029
Inventory · days	2.7 · 178	vs ~135 a year ago

Micron bought back heavily in 2018 and 2022 at its cycle tops; the stock halved each time. Peak cash flow is being committed to shares at peak multiples and to fabs that arrive after the crossover.

The contract book: 10 "New Business Model" agreements (5 Aug 2026)

Term	Disclosure
Agreements / customers	10 NBM agreements with 8 customers (hyperscalers, OEMs)
Minimum revenue	\$93.9bn "assuming floor pricing"
Duration	Up to 5 years; weighted average over 4 years → FY27-FY31
Bit coverage	>50% of FY27 bits; ~2/3 of FY28 bits (Citi, 8 Sep: "50% by end-FY28")
Margin at floor	"around 80% gross margin, even at floor pricing"
Guarantees	\$16.5bn of customer financial guarantees; \$2.7bn of it on SanDisk's balance sheet
Already re-opened	2 of 8 customers (one extended, one added volume)
Pricing above floor	Floors plus market-linked uplift; the uncontracted third of bits is at spot



Floor vs consensus, per bit

Floor revenue per year (\$93.9bn ÷ 5) for ~2/3 of FY28 bits	\$18.8bn
Floor price applied to 100% of bits	~\$28bn
FY28 consensus revenue per bit ÷ floor price per bit	~2.0×
If covered bits earn only the floor, the uncovered third must earn	~4× floor

The contracts are real for four years. The pitch is not that the floors break; it is that the price already assumes something **better than the floors, permanently** (slide 5).

The bull case in the bulls' own words, and none of it changes what the price requires after 2031

Demand changed in kind; supply is capped to 2028

"We are unlikely to see any significant increase in incremental supply through 2028."

Samsung, Q2 2026 call, 30 Jul 2026

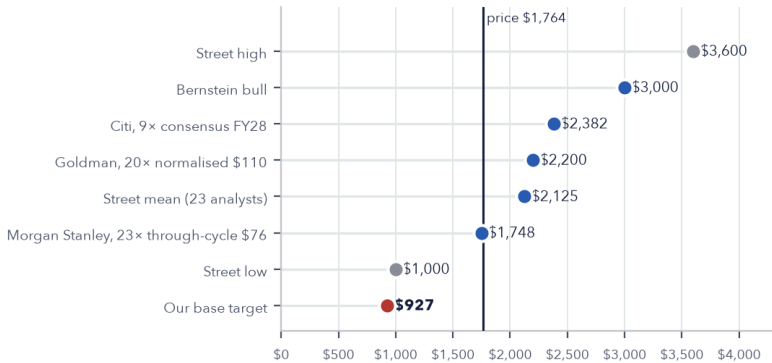
Bits are "on allocation beyond calendar 2027."

SanDisk management, FQ4 call, 5 Aug 2026

- **Enterprise SSD went from 26% to 48% of NAND bits in a year.** Nvidia's Vera Rubin rack specifies 16 TB of NAND per GPU; HDD is sold out into FY28.
- **Hyperscaler capex ~\$775-800bn in 2026**, all five signalling higher 2027.
- **No greenfield NAND wafers before 2H28**; Samsung and SK hynix cut NAND wafer starts to fund HBM; 2026 NAND capex +5% vs DRAM +14%.
- **Contracts everywhere:** SanDisk \$93.9bn floors; Micron \$100bn take-or-pay; Samsung 60-70% of capacity under 3-year-plus deals.

Management's FY28-30 model (Investor Day, 13 Aug)

Mid/high-teens revenue growth · ~80% gross margin · **~75% operating margin** · ~50% FCF margin · 100% of excess cash returned.



"The stock is already priced for a cliff"

8.2x

FY27 consensus P/E (6.7x FY28). "Cheap" is the whole bull valuation argument.

9x

Citi's multiple "like a mature cyclical", which still clears spot on consensus FY28

\$2,125

Mean of 23 sell-side targets (\$1,000-3,600); 20 of 24 ratings are Buy, 1 Sell

\$214

Bernstein FY30 EPS in a floor-stress case (\$0.11/GB on uncovered bits): price is ~8x a "protected trough"

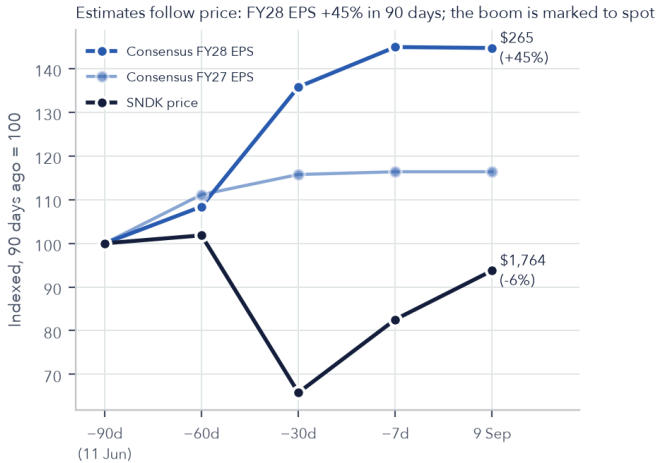
Who owns it

Institutional ownership	81%
Hedge funds (Q2-26 13F) · value held	128 · \$25.6bn
Short interest (FINRA 14 Aug) · days to cover	5.3% · ~1
Index events	MSCI 31 Aug · S&P 100 21 Sep
Company buyback, FQ4 · remaining	\$4.5bn · \$15.5bn

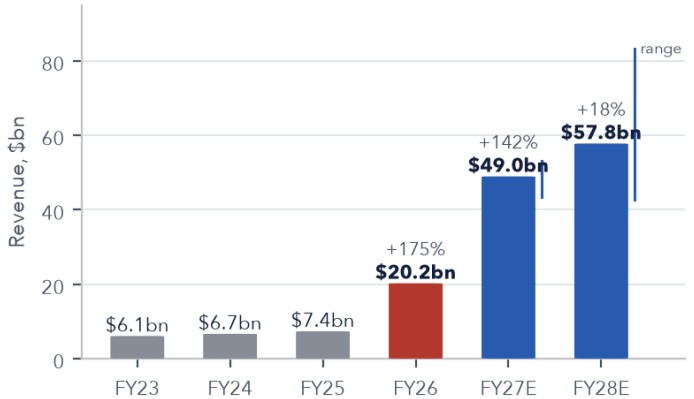
The market is not naïve about the cycle: an 8x multiple is the market pricing the boom's end. The debate is the **terminal value**, ~70% of the enterprise value, and that is where we disagree.

Who makes NAND. Samsung, SK hynix + Solidigm, the Kioxia + SanDisk JV, Micron, and YMTC (14% bit share). In 2026 Samsung and SK hynix diverted wafer starts to HBM, the JV moved to BiCS10, Micron's greenfield fab lands 2H28, and YMTC pulled Wuhan Fab 3 into 2H26 (slide 7).

Estimates follow price: the boom is marked to spot

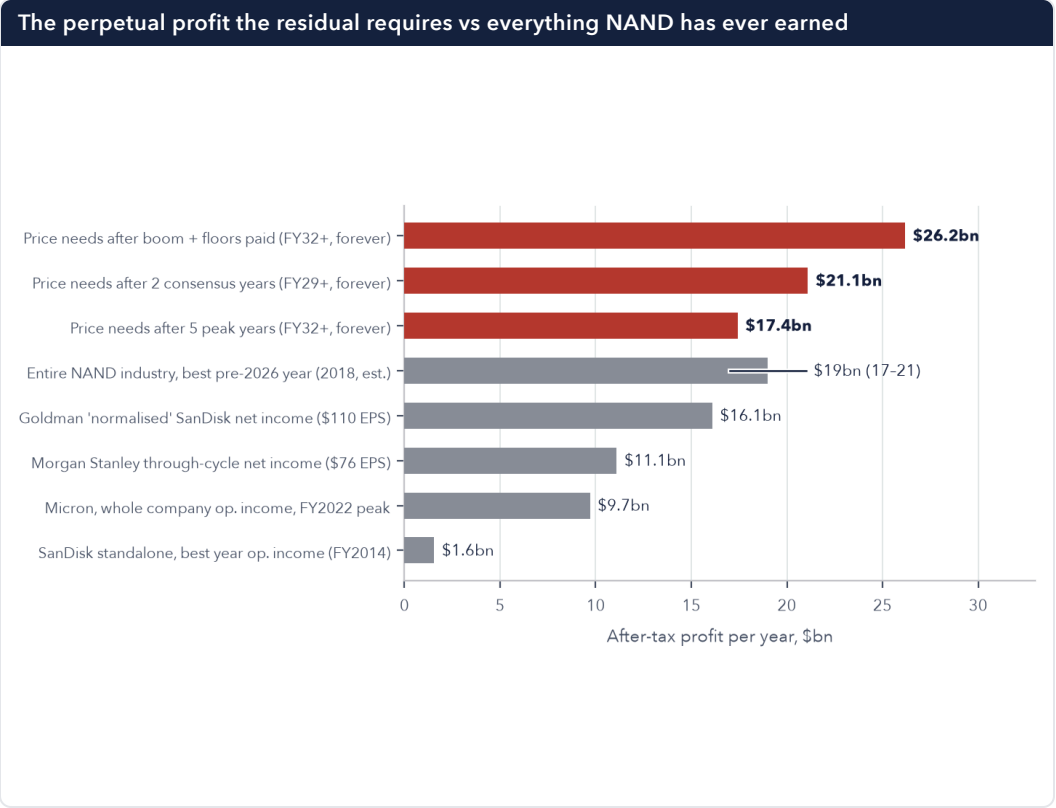
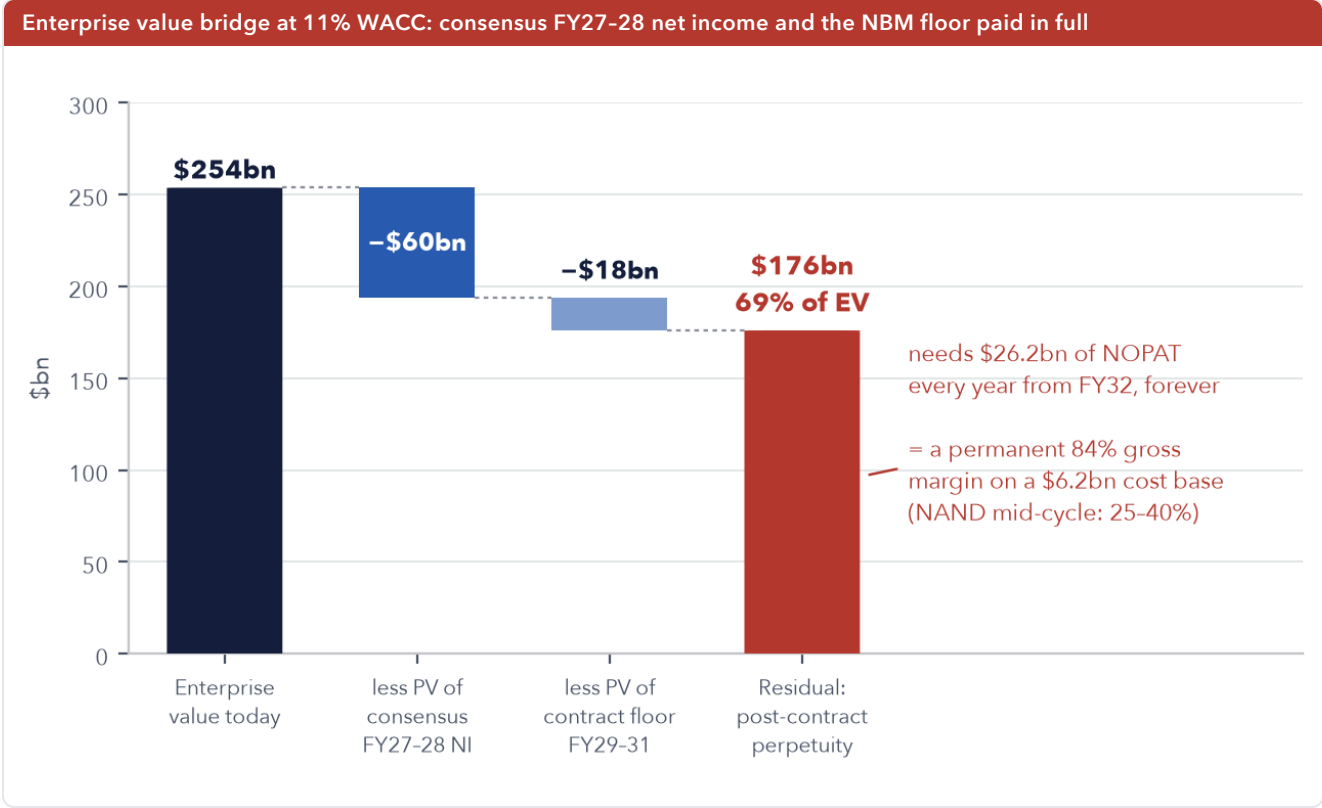


Consensus revenue: three flat years, then 8x in three (bits +15% a year)



Consensus FY28 EPS moved from \$183 to \$265 (+45%) in 90 days while the stock round-tripped \$1,882 → \$1,238 → \$1,764. Morgan Stanley's "through-cycle" EPS went \$30 → \$48 → \$76 in four months. FY27-28 numbers are derived from spot; "normalised" numbers are derived from spot; the price is validated by numbers derived from the price. EPS-upgrade breadth has fallen from 92% to 77%.

Pay the Street in full, then pay the contracts in full: the residual still needs \$26.2bn a year, forever

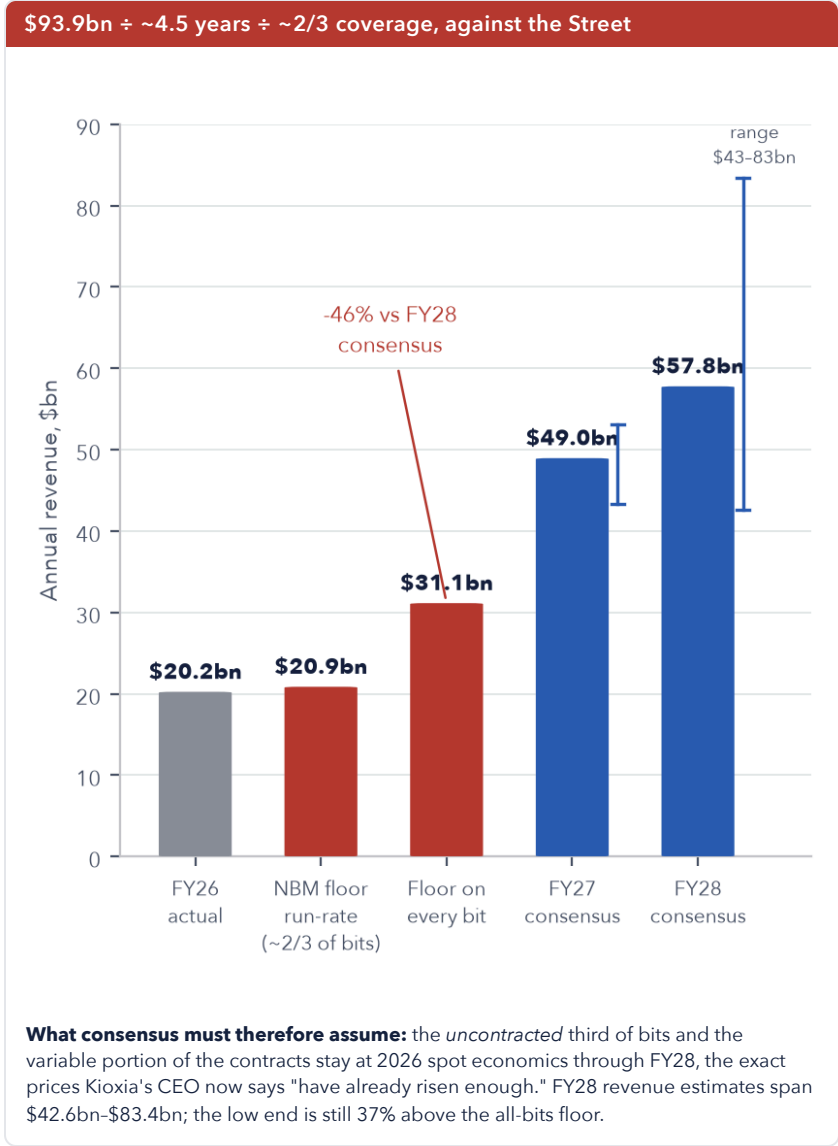


Three ways to read the same enterprise value of \$254bn					
What you pay for	PV at 11%	Share of EV	Residual EV	Perpetual NOPAT the residual needs	
A. Consensus FY27-28 net income (\$31.3bn, \$38.8bn)	\$60bn	24%	\$194bn	Row A: grant management's entire long-term model for five years, 75% operating margins on \$58bn of revenue, and you have paid for 54% of the enterprise value. The rest is a claim on FY2032 onward that needs \$17.4bn of after-tax profit every year, indefinitely.	
B. Five full peak years FY27-31: FY28 consensus held flat for the whole contract life (\$187bn of net income)	\$136.6bn	54%	\$117bn	The entire NAND industry, all five producers, earned an estimated \$17-21bn of operating profit in its best pre-2026 year (2018) and \$12-15bn in 2021-22. Goldman's "normalised" SanDisk (\$110 EPS = \$16bn) is below what the price needs as a floor for eternity. Terminal growth 2%, tax 15% (guided), 11% WACC (10-13% band on A1).	
C. Consensus FY27-28 + NBM floor (\$18.8bn/yr at 80% GM) FY29-31, uncontracted bits at 40% GM	\$78bn	31%	\$176bn		

Floor pricing on every bit is \$31.1bn a year, -46% below FY28 consensus, and the "\$16.5bn of guarantees" is \$2.7bn of cash

SHORT SNDK

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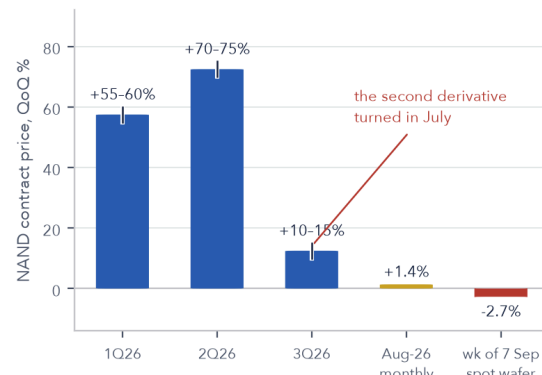


Consensus FY27 needs another doubling of blended price; the second derivative turned in July; every forecaster dates the crossover 2H27

SHORT SNDK

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NAND contract pricing is decelerating on schedule



Morgan Stanley's memory analyst (21 Jul) calls a **4Q26 contract-price peak**. Spot 512Gb wafers fell 2.7% in the week of 7 Sep with "suppliers lowering quotes"; retail \$/TB has been flat since February.

Demand destruction is already in the numbers

-32%

SanDisk Consumer revenue, QoQ, FQ4-26: "some impact on the TAM itself" (CFO)

-11%

2026 PC units, IDC; client-SSD OEM inventories "elevated" after the H1 pre-build

"Prices have already risen enough."

Kioxia CEO, SanDisk's JV partner, Bloomberg, 8 Sep 2026

-16.7%

2026 smartphone units, IDC (26 Aug), the steepest decline on record; memory costs up 300%+

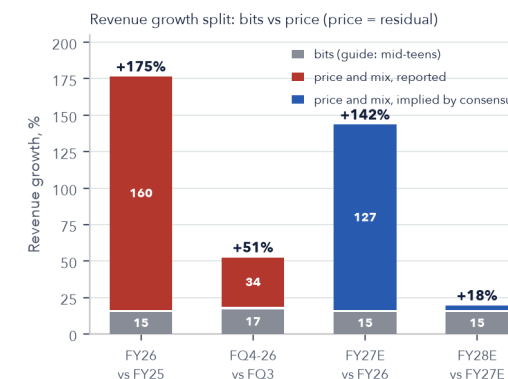
+1.4%

August monthly NAND contract benchmark, after +55-60%, +70-75% and +10-15% quarters

Consumer buyers are at "the limit of price tolerance"; wafer demand has "weakened so significantly."

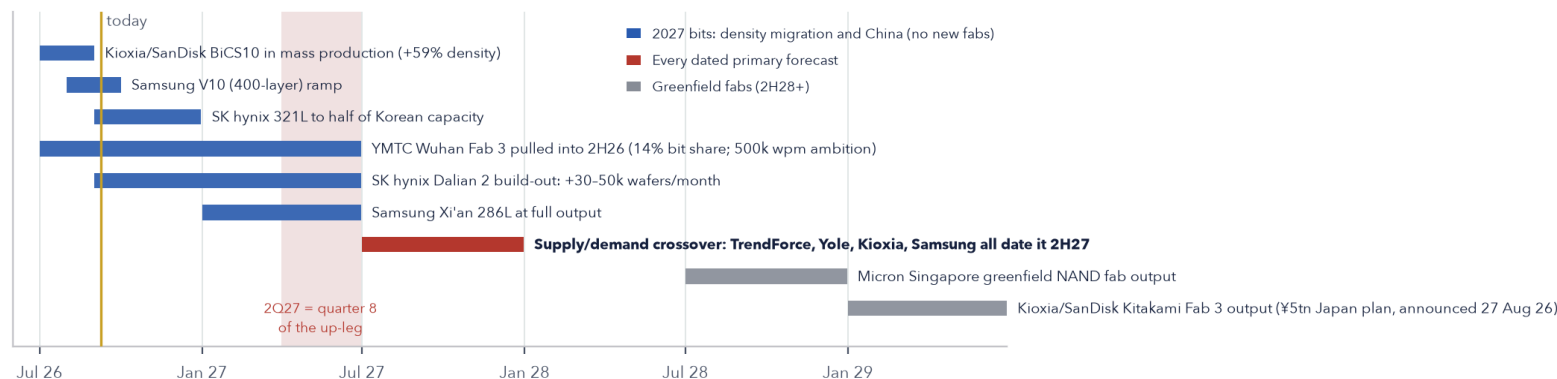
TrendForce, 3Q26 NAND contract release

Revenue is price: FY27 consensus needs another doubling



Bits from the company's "mid-teens sellable bit growth" guide; price and mix is the residual. When price stops rising, growth stops; when it falls, operating leverage runs in reverse against a cost base SanDisk cannot cut.

Supply roadmap: 2027 bits come from density and China, not fabs, and every dated forecast puts the crossover in 2H27



BiCS10 delivers +59% bits per wafer, Samsung V10 and SK hynix 321L similar; YMTc has 14% bit share and pulled Fab 3 into 2H26; Dalian 2 adds 30-50k wafers a month. Then the fabs: the ¥5tn Kioxia/SanDisk plan (27 Aug), Micron Singapore 2H28. Peak profits fund the next glut, every cycle.

The crossover, and the base rates of the down-leg

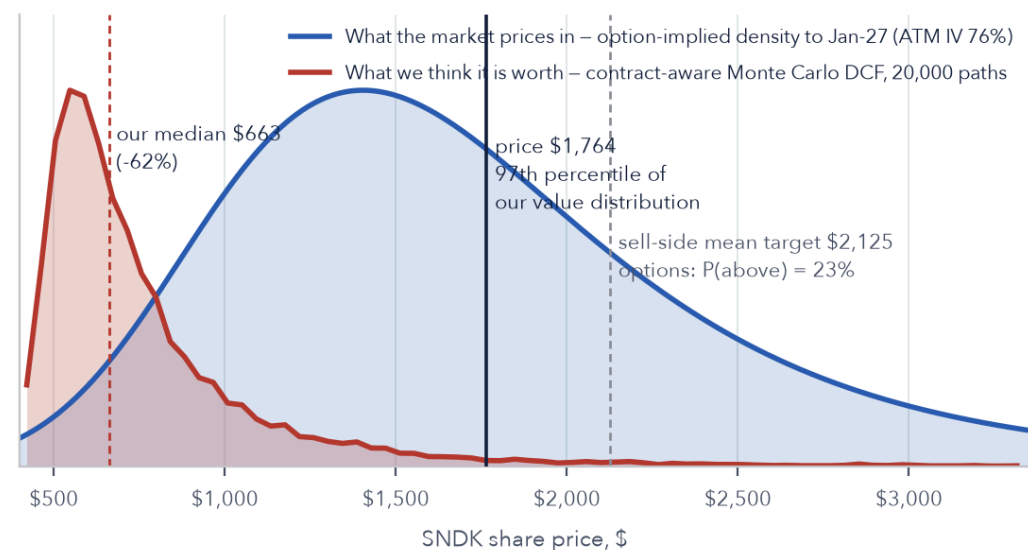
Source	Crossover	On pricing
TrendForce (21, 30 Jul)	2H27	"increasing downward pressure"
Yole Group	2027	"beginning of a down cycle"
Samsung (30 Jul)	2H27	supply "normalising"
Kioxia (27 Aug, 8 Sep)	2027 conversions	"prices have risen enough"

- No NAND up-leg has run much past eight quarters.** Prices stabilised in 2Q25; 2Q27 is quarter eight.
- In 2018 and 2022 the decline began **within one quarter** of the crossover; peak-to-trough took 19-24 months; wafer prices fell 70% in 2018-19 and 30-35% in one quarter in 3Q22.

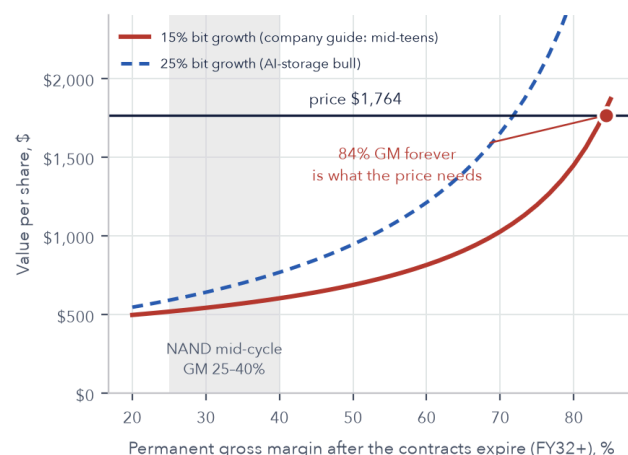
The principal risk is timing: 1H27 or 2H27. Both are inside the window, and the stock leads the print (slide 9).

Our value distribution has a median of \$663; today's price sits at its 97th percentile

What we think it is worth vs what the market prices in, on one price axis



Sum-of-parts value by the one assumption that matters



Post-contract GM	30%	40%	50%	60%	70%	80%
Value / share	\$541	\$602	\$687	\$814	\$1,025	\$1,448

Method. FY27-28 at consensus; FY29-31 NBM floor at 80% GM plus uncontracted bits at mid-cycle; FY32+ everything at mid-cycle. Cost base grows with bits (15%) less cost-per-bit declines (12%); WACC 11%; g 2%.

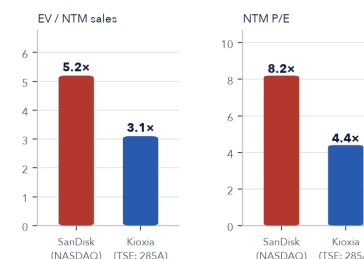
Monte Carlo (20,000 paths). Post-contract GM drawn N(45%, 15%), above every mid-cycle in NAND history; bit growth N(18%, 7%), above guidance; WACC 10-13%; consensus hit/miss $\pm 12\%$; 20% chance floors are renegotiated. Result: p5-p95 \$460-\$1,463, **P(value > price) = 2.8%**. Re-centre GM at 60% and the price is still at the 88th percentile.

Options. Risk-neutral, not a forecast: P(above \$1,764 in Jan) 39%; P(reach \$2,125) 23%; P(below \$1,000) 14%; 25Δ risk-reversal +4.2%, calls are the expensive side.

Thesis 3 · The bulls' normalised numbers, and same-fab relative value

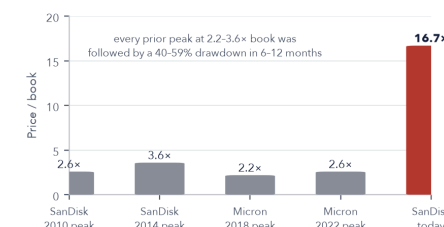
Framework	Norm. EPS	Net income	Multiple at \$1,764	Their multiple → target
Morgan Stanley through-cycle	\$76	\$11.1bn	23.2×	23× → \$1,748 ≈ price
Goldman normalised	\$110	\$16.1bn	16.0×	20× → \$2,200
Bernstein FY30 floor stress	\$214	\$31.3bn	8.2×	needs 60% coverage at \$0.29/GB and half the shares retired
What the price needs after five peak years		– \$17.4bn/yr	forever	≈ the whole NAND industry's best year

Memory has never sustained 20×+ on normalised earnings: Micron traded 4-6× forward EPS through the 2018-19 rollover. Across 33 documented NAND company-years the median operating margin is 9%, the best three-year run 24%, and 10 of 33 lost money (A2).



Same fabs, different price

Kioxia co-owns Flash Ventures, is ~20% larger by revenue, makes no HBM either, and trades at 3.1× forward sales and 4.4× forward earnings vs SanDisk's 5.2× and 8.2×. **SanDisk at Kioxia's multiples is \$940-\$1,080** (-39% to -47%), a cycle-agnostic cross-check. The gap is a listing venue and an index flow, not a business.



Price-to-book **16.7×** vs 2.2-3.6× at every prior SanDisk and Micron cycle peak, each followed by a 40-59% drawdown within 6-12 months (A2). Book value is the one denominator that is not marked to spot.

Base target \$927 (-47%) on a cut FY28 at a cyclical multiple; +28% expected return at 4.3 : 1

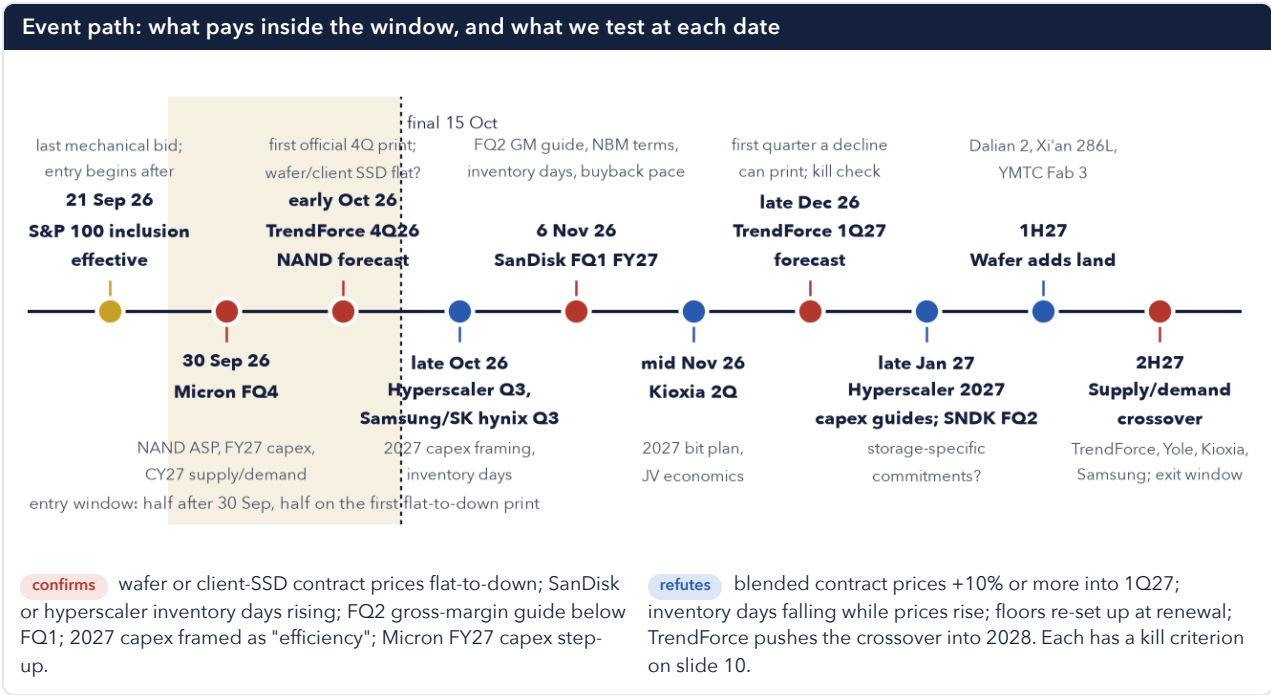
Twelve-month scenarios: what the market pays once the terminal narrative breaks (multiple × FY28 EPS)

Scenario	Weight	FY28 revenue	Op. margin	FY28 EPS	Multiple	Target	Short return	Why this multiple
Base: contract prices flat by 1Q27, falling from 2Q27	50%	\$38bn	60%	\$132	7×	\$927	+47%	Consensus cut ~35%; floors cushion margin; 7× = Micron 2018-19 rollover range
Deep bust: bullwhip unwind + China supply; floors partly renegotiated	20%	\$30bn	45%	\$78	8×	\$627	+64%	Revenue back to the FQ4-26 run-rate; higher multiple on trough EPS
Against us: prices rise through CY27; FY28 consensus holds	30%	\$57.8bn	75%	\$252	9×	\$2,264	-28%	Management's 75% OM on full consensus at Citi's "mature cyclical" 9×

Probability-weighted short return +28% · reward/risk 4.3 : 1. The tail beyond the against-us case (June high \$2,354; Bernstein \$3,000) is handled by sizing and the stop.

Intrinsic value (slide 8) is below every target: targets are what the market pays at a rollover, not DCF value.

Scenario	Weight	Target (\$)	Return (%)
Against us (30%)	30%	\$2,264	-28%
Base (50%)	50%	\$927	+47%
Deep bust (20%)	20%	\$627	+64%



Trade construction

Size · borrow	79% idiosyncratic vol → ~2.5% of NAV notional per 1% of NAV at risk per quarter; 2.5-3% is full-conviction size. Borrow 0.28% GC, 2.1m shares available (IBKR, 9 Sep), no dividend.
Entry	Half after the 21 Sep inclusion print and Micron on 30 Sep; second half on the first TrendForce print showing wafer / client-SSD contract prices flat-to-down (early Oct for 4Q26, or late Dec for 1Q27).
Hedge	Half the SMH beta (β 1.73, R^2 43%) $\approx$ \$0.85 long SMH per \$1 short. A full hedge buys back Micron and Samsung, which fall in the payoff state; no hedge leaves 43% of variance as sector beta.
Stop / tail	Close above the June high \$2,354 = re-rating, not testing: cover half, reassess. No call protection: a rolled 3-month 25 Δ call at 76% IV costs ~5% of notional per quarter, the entire expected return.
Expected path	Down 10-20% into November. The 6 Nov print will likely show 83-85% GM and top-of-range revenue. We hold because the print cannot change the post-2031 arithmetic.
Exit	First two negative quarterly contract prints, which base rates put in 2Q-3Q27; the six-month minimum is met by construction. The pair a fund would run: short SNDK / long Kioxia (outside the mandate).

P&L map, 2.5% of NAV short, half-beta hedged

	Bust (20%)	Base (50%)	Against (30%)	Stop \$2,354	Expected
NAV impact	+1.6%	+1.2%	-0.7%	-0.8%	+0.7%

What breaks the short, what we do about it, and the four pre-committed conditions under which we cover

The other side of the trade, and our mitigants																	
Risk	How it hurts	Mitigant / why it is already in the numbers															
1. Prices keep rising through CY27 (allocation "beyond 2027"; HBM diverts more wafers; hyperscaler capex > \$1tn)	FY28 consensus holds or rises; stock re-rates on "structural"; we are down 28% in the against-us case, more in the tail	Weighted 30% at 9× full consensus. We pay consensus in full and the price <i>still</i> needs \$17.4bn forever. Staged entry; stop at \$2,354; sized to a 1% NAV loss per quarter															
2. The floors hold and the market believes them	Trough earnings are cushioned; "8× a protected trough" persists	Floors on every bit = \$31.1bn/yr, -46% vs consensus; the base case already assumes 60% op margin in FY28. The price needs more than the floors															
3. Flows, momentum, buyback (S&P 100 21 Sep; \$15.5bn authorisation; 128 hedge funds)	The stock grinds higher on no news at 76% IV	Entry <i>after</i> inclusion; half position until the first flat print; the buyback retires ~6% of shares at ~\$1,700; Micron bought back at the top in 2018 and 2022 and halved anyway															
4. Structural change: HBF, AI-storage TAM (High-Bandwidth Flash; 16 TB NAND per GPU)	NAND becomes an AI bottleneck with pricing power that survives the cycle	HBF is pre-revenue until 2028 and excluded from every bull model; Vera Rubin demand is in the consensus we pay. Structural pricing power shows up as floors re-set up at renewal: kill criterion 3															
5. Squeeze / borrow	Short interest rises, borrow tightens	5.3% of float, ~1 day to cover, 0.28% GC, 2.1m shares available. The crowd is long, not short															
6. Timing wrong by two quarters	Crossover in 2028; carry is trivial, drawdown and opportunity cost are not	The stock leads the print by 1-2 quarters; the 6-month minimum is met; kill criterion 1 forces a re-underwrite by the 2Q27 print															
7. The contracts get longer (new NBMs to FY33+)	The perpetuity starts to look contracted	Kill criterion 3 covers it: coverage above 80% of FY29 bits or floors stepping up → exit															
Everything in the "How it hurts" column is a price path, not a change in the post-2031 arithmetic. Only the kill criteria describe changes in the arithmetic.																	
The asymmetry		Refreshed before the final															
Probability-weighted 12-month short return	+28%	<table><tr><th>Date</th><th>Input</th><th>Slides</th></tr><tr><td>21 Sep</td><td>S&P 100 inclusion print</td><td>2, 9</td></tr><tr><td>30 Sep</td><td>Micron FQ4: NAND ASP, FY27 capex</td><td>7, 9</td></tr><tr><td>early Oct</td><td>TrendForce 4Q26 NAND contract forecast</td><td>7, 9, 10</td></tr><tr><td>14 Oct</td><td>Engine re-run on refreshed prices, consensus, options</td><td>2, 5, 8, 9</td></tr></table>	Date	Input	Slides	21 Sep	S&P 100 inclusion print	2, 9	30 Sep	Micron FQ4: NAND ASP, FY27 capex	7, 9	early Oct	TrendForce 4Q26 NAND contract forecast	7, 9, 10	14 Oct	Engine re-run on refreshed prices, consensus, options	2, 5, 8, 9
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14 Oct	Engine re-run on refreshed prices, consensus, options	2, 5, 8, 9															
Reward / risk, against-us case at 30%	4.3 : 1																
Share of 20,000 Monte Carlo paths reaching today's price	2.8%																
Loss on a 2.5% position if the stop is hit	−0.8% of NAV																
Carry (borrow, no dividend)	0.28% p.a.																

Risk map and kill criteria

- 1 TrendForce 1Q27 and 2Q27 prints both show blended NAND contract prices still rising QoQ → timing wrong; cover half. late Dec 26 · late Mar 27
- 2 Hyperscaler 2027 capex guides accelerate above ~\$1tn with storage-specific commitments, and SanDisk FQ2 shows inventory days falling → cover half. late Jan 27
- 3 SanDisk discloses floor prices ≥ 2Q26 ASP with guarantees stepping up, or new NBMs taking coverage above 80% of FY29 bits → the terminal is being contracted; exit. 6 Nov 26 · Jan 27
- 4 Close above \$2,354 → re-rating, not testing; cover half. any day

What would make us wrong: a structural, not cyclical, change in NAND pricing. Each version of it has a date above. Everything else, a strong FQ1, a rally into November, a higher June high, is priced, sized, or stopped.

The engine behind slides 5 and 8: consensus paid directly, the floor paid directly, the residual solved for

Cash-flow build, sum-of-parts case C at 40% post-contract gross margin (\$bn, FY Jun)							
Year	Regime	Revenue	GM	Cash COGS	Opex	NOPAT	FCF
FY27	consensus	49.0	–	5.8	2.12	31.3	27.0
FY28	consensus	57.8	–	5.9	2.23	38.8	37.4
FY29	floor + mid-cycle	22.1	73%	6.0	2.34	11.7	11.7
FY30	floor + mid-cycle	22.1	73%	6.1	2.45	11.6	11.6
FY31	floor + mid-cycle	22.1	72%	6.1	2.58	11.4	11.4
FY32	mid-cycle 40%	10.3	40%	6.2	2.71	1.2	1.2
FY33	mid-cycle 40%	10.5	40%	6.3	2.84	1.1	1.1
FY34	mid-cycle 40%	10.6	40%	6.3	2.98	1.1	1.0
FY35	mid-cycle 40%	10.7	40%	6.4	3.13	1.0	1.0
FY36	mid-cycle 40%	10.8	40%	6.5	3.29	0.9	0.9
Terminal value = FY36 NOPAT × 1.02 / (11% – 2%), discounted 10 years. Equity = EV – debt \$0.2bn + cash \$4.8bn over 146.4m shares → \$602 at 40% GM. FY29–31 revenue = floor \$18.8bn plus uncontracted bits (1/3 of the cost base) at mid-cycle. Cash COGS starts at FY26's \$5.8bn, grows with bits (+15%) less cost-per-bit (–12%); opex \$2.12bn growing 5%; incremental capital 15% of revenue growth; tax 15%.							
Case A, step by step							
Step							\$bn
PV of FY27 consensus net income: 31.3 / 1.11							28.2
PV of FY28 consensus net income: 38.8 / 1.11 ²							31.5
Enterprise value 253.7 less both							194.0
× 1.11 ² × (0.11 – 0.02) / 1.02 = perpetual NOPAT from FY29							21.1
Verification tests (analysis/tests/test_engine.py, 17 passing)							
Group	What is asserted						
DCF engine (5)	Zero-growth EV = NOPAT/WACC; reverse solvers round-trip; terminal NOPAT round-trips; forecast period monotone						
Options (5)	Put-call parity; IV round-trips; sub-intrinsic quotes rejected; B-L recovers a lognormal; smile fit recovers flat vol						
Monte Carlo (3)	Degenerate draws match the deterministic value; correlation sign propagates; percentiles correct						
Contract model (4)	Required NOPAT reproduces EV; required GM reproduces NOPAT; monotone in margin and growth; case-B closed form						

Source: analysis/sndk_contract_dcf.py and analysis/quant/dcf.py (Rappaport-Mauboussin FCFF engine); analysis/tests/test_engine.py.

Residual algebra and assumptions

The required perpetual NOPAT

Residual = EV – ∑ PV(FCF_t) for t ≤ T. If the post-T business earns constant NOPAT growing at inflation i with value-neutral reinvestment: Residual = NOPAT × (1+i) / (WACC – i) / (1+WACC)^T, so **NOPAT_{req} = Residual × (1+WACC)^T × (WACC – i) / (1+i)**. Required gross margin: GM_{req} = 1 – COGS₃₂ / (NOPAT_{req}/(1–τ) + Opex₃₂ + COGS₃₂).

Assumption	Base	Provenance
WACC	11%	rf 4.8% + β 1.7-2.2 × ERP 4.5% → 11-13%; low end used
Terminal growth	2%	Inflation; bit growth is in the cost base, not the terminal
Tax	15%	Guided non-GAAP rate
Consensus FY27 / FY28 NI	\$31.3bn / \$38.8bn	EPS \$214 / \$265 × 146.4m shares (Yahoo/LSEG)
NBM floor / yr · GM at floor	\$18.8bn · 80%	\$93.9bn ÷ 5 years; FQ4 call
Bit growth · cost/bit	+15% · –12%	"Mid-teens" guide; nodal productivity
Shares	146.4m basic	Diluted ~157m would lower per-share values ~7%

Required perpetual NOPAT across the WACC band (case A)

	WACC	10%	11%	12%	13%
NOPAT/yr from FY29		\$18.3bn	\$21.1bn	\$24.0bn	\$26.9bn

A lower discount rate makes the required perpetual profit smaller but still ~12× SanDisk's best pre-2026 year and about the whole industry's 2018 profit pool.

Assumptions that favour the bull case

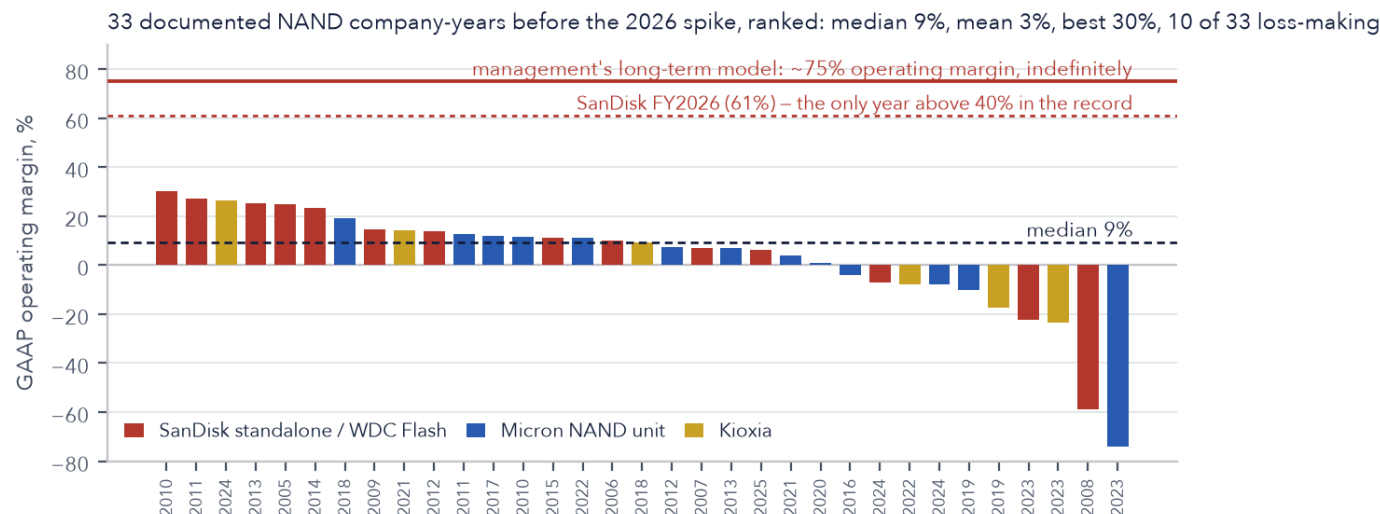
- Consensus FY27-28 net income is paid in full with no reinvestment charge beyond 15% of revenue growth; FY29-31 floor years use the 80% gross margin management quoted *at floor*.
- WACC 11% is the low end of a CAPM band for a 1.7-2.7 beta stock; cost per bit falls 12% a year in perpetuity; tax is the guided 15%, not the 21% statutory.
- Basic shares (146.4m) are used; the ~157m diluted count would cut every per-share value by ~7%.
- No value is deducted for the \$6.6bn of JV commitments or the ¥5tn Japan plan.

33 documented NAND company-years: median operating margin 9%, best year 30%, best three-year run 24%, 10 loss-making years

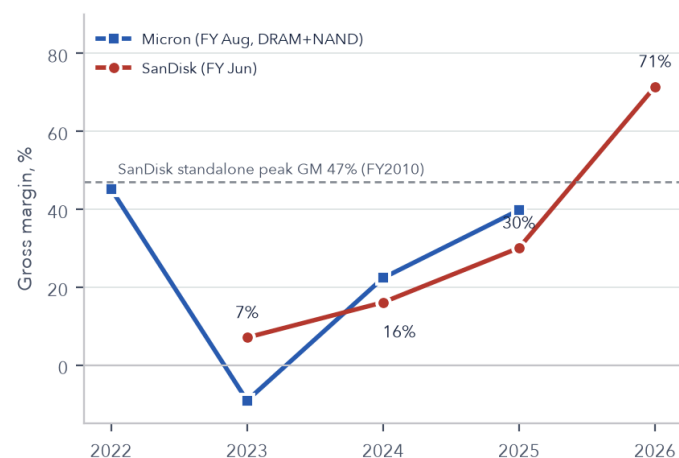
APPENDIX · for Q&A, not presented

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Every documented company-year, ranked (GAAP operating margin)



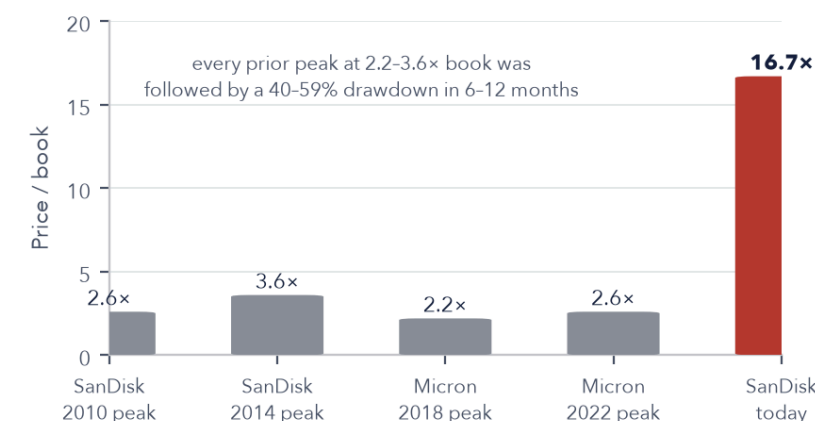
Annual gross margin, SanDisk vs Micron: gross losses to 71% in three fiscal years



The industry's operating profit pool: ~\$17-21bn in 2018 (all producers, estimated from segment disclosures), ~\$12-15bn in 2021-22. SanDisk's FY26 operating income alone (\$12.5bn) equals the whole industry's 2021-22 peak; the perpetual profit the price needs (\$17.4bn-\$26.2bn) equals or exceeds the whole industry's best year. The required permanent gross margin (84%) is above every number in the record except the current quarter.

Cycle table and the record after peaks

Peak	P/B at peak	What followed
SanDisk, 2010	2.6×	–45% within 12 months as MLC pricing fell
SanDisk, 2014	3.6×	–59% by 2015; sold to WDC 2016
Micron, May 2018	2.2×	–50% in 7 months; earnings peaked two quarters later
Micron, Jan 2022	2.6×	–45% by Sep 2022; loss-making by 2023
SanDisk, today	16.7×	–



Company	Operating margin by year, %
Kioxia	2018 +9%, 2019 -18%, 2021 +14%, 2022 -8%, 2023 -24%, 2024 +26%
Micron NAND unit	2010 +11%, 2011 +13%, 2012 +7%, 2013 +7%, 2016 -4%, 2017 +12%, 2018 +19%, 2019 -10%, 2020 +1%, 2021 +4%, 2022 +11%, 2023 -74%, 2024 -8%
SanDisk standalone / WDC Flash	2005 +25%, 2006 +10%, 2007 +7%, 2008 -59%, 2009 +15%, 2010 +30%, 2011 +27%, 2012 +14%, 2013 +25%, 2014 +24%, 2015 +11%, 2023 -22%, 2024 -7%, 2025 +6%

Excluded as estimates or boom-affected: WDC Flash 2020-22 (~+4/+12/+20, est.) and Kioxia FY25 (+37, includes the 2026 spike). Kioxia FY2020 and Samsung / SK hynix NAND-only margins are not disclosed. Micron's NAND unit never exceeded 19% in thirteen disclosed years.

Inputs centred above NAND history and above guidance, and the price still sits at the 97th percentile

Value per share across the two assumptions that matter (row: permanent post-contract gross margin); red below \$1,764, blue above										
GM ↓ · bit growth →	10%	15%	20%	25%	30%	GM ↓ · WACC →	10%	11%	12%	13%
30%	\$514	\$541	\$581	\$640	\$724	30%	\$549	\$541	\$533	\$525
40%	\$555	\$602	\$669	\$767	\$905	40%	\$620	\$602	\$586	\$571
50%	\$613	\$687	\$792	\$944	\$1,159	50%	\$718	\$687	\$660	\$636
60%	\$699	\$814	\$977	\$1,210	\$1,536	60%	\$866	\$814	\$770	\$733
70%	\$844	\$1,025	\$1,285	\$1,651	\$2,165	70%	\$1,112	\$1,025	\$955	\$896
80%	\$1,132	\$1,448	\$1,896	\$2,532	\$3,423	80%	\$1,604	\$1,448	\$1,322	\$1,219

To reach \$1,764 you need **either** a permanent 80% gross margin with 20%+ bit growth, **or** a 70% margin with 30% bit growth, every year after the contracts expire. No cell at 60% margin or below reaches the price at any WACC or growth rate. The consensus FY27-28 years and the FY29-31 floor are paid in full in every cell.



Provenance of every distribution		
Input	Distribution	Why it favours the bull case
Post-contract GM	N(45%, 15%), clipped 5-85%	NAND mid-cycle history 25-40%; 2026 peak 85%; centre is above every non-boom year
Bit growth	N(18%, 7%), clipped 0-40%	Company guides mid-teens; AI-storage bulls 25-30%
Cost/bit decline	N(12%, 4%)	Nodal cadence; higher declines raise value
WACC	Triangular 10 / 11.5 / 13%	CAPM band; mode below midpoint
FY27-28 vs consensus	N(100%, 12%)	Symmetric hit/miss on the Street's boom
Floor renegotiation	20% chance × U(60-90%) of floor revenue	Enforceability untested; 80% of paths keep the floors whole

Sensitivity to the margin centre			
Post-contract GM centre	Median value	Price percentile	P(value > price)
45% (base; above NAND history)	\$663	97th	2.8%
60% (5,000 paths, all else equal)	\$883	88th	12%

A 60% permanent gross margin is 13 points above the best year NAND had before 2026 and 30 points above its best three-year run; even then, roughly one path in eight reaches today's price. The distribution is not the argument; slide 5's arithmetic is. The distribution shows how much room the argument has.

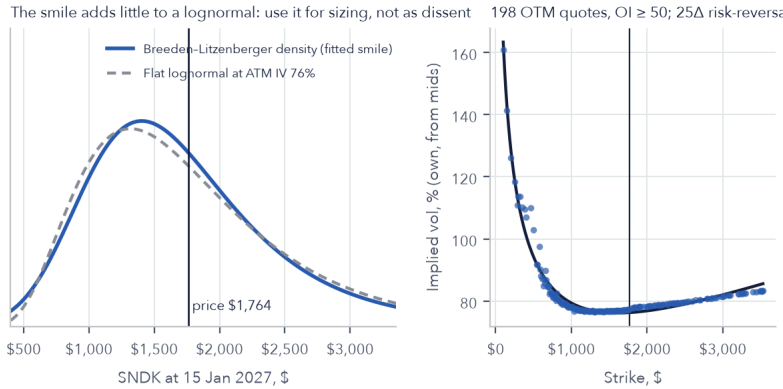
The option-implied density is what the market prices, not what we predict; the position is majority idiosyncratic; the crowd is long

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Breeden-Litzenberger pipeline and validation (Jan-2027 expiry)

The smile adds little to a lognormal: use it for sizing, not as dissent



price \$1,764

SNDK at 15 Jan 2027, \$

Implied vol, % (own, from mids)

Strike, \$

Pipeline. Mid quotes, bid > 0, OI ≥ 50, OTM only (198 quotes) → our own Black-Scholes implied vols → smoothing spline in vol space vs log-moneyness → reprice calls on a 4,001-point grid → second difference → density ≥ 0, ∫ = 0.96 before renormalisation over the quoted strike range → tight call-spread digital check within 2 points.

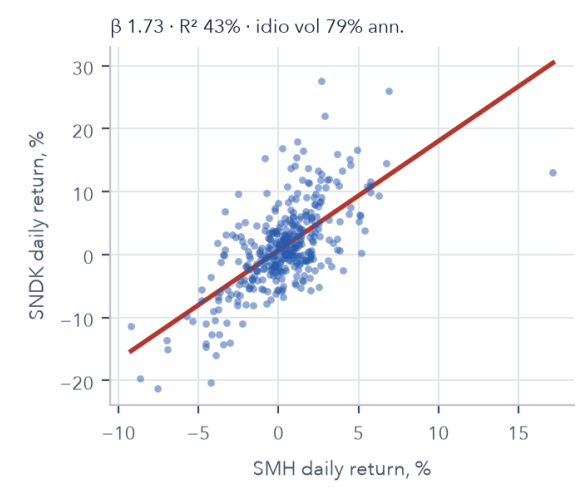
Risk-neutral, not real-world. It embeds risk premia, which is why we use it as "what the market prices" and as a sizing input, never as a forecast. The smile sits within ~5 points of a flat lognormal at 76% IV, and its tails moved 15 points in a week of data.

What the market prices, Jan-27 and Mar-27

Probability (risk-neutral)	Jan-27	Mar-27
Above \$1,764 (today's price)	39%	35%
Above \$2,125 (mean sell-side target)	23%	22%
Above \$2,354 (June high; our stop)	15%	16%
Below \$1,200	25%	33%
Below \$1,000	14%	21%
ATM implied vol · 25Δ risk-reversal	76% · +4.2%	78% · +4.4%

Calls are the rich side of the smile: the market is paying up for upside, not downside, consistent with a long-crowded, uncrowded-short register. Horizon mismatch: sell-side targets are 12-month; the density is 4-6 months. A week earlier the same pipeline priced P(below \$1,000) at 31%; it is now 14%. The density moves with the smile, which is why it sizes the trade and does not make the argument.

Factor exposure: what we are actually underwriting



β 1.73 · R² 43% · idio vol 79% ann.

SNDK daily return, %

SMH daily return, %

vs	β	R²	Idio vol
SMH (semis)	1.73	43%	79%
QQQ	2.67	33%	86%

Daily returns since Mar-25, OLS with Newey-West (6 lags) errors. 57% of variance is idiosyncratic: the trade is a NAND-pricing bet, not a semis bet. R² of 30-60% is normal for a single stock; we do not claim historical alpha. Half-beta hedge in SMH keeps the payoff-state exposure (Micron, Samsung fall with NAND) while removing the pure index leg.

The IV cone: why the position is 2.5% of NAV, not 10%

Horizon at 76% ATM vol	1 month	3 months	6 months
One-standard-deviation move	±22%	±38%	±54%
Loss on 2.5% notional at 1σ against us	0.5% NAV	1.0% NAV	1.4% NAV

Positioning and the flow calendar

Item	Value	Note
Short interest (FINRA, 14 Aug settlement)	7.68m sh · 5.3%	Yahoo's 8.2% is an artefact; ~1 day to cover
Borrow · availability	0.28% · 2.1m sh	General collateral; IBKR 9 Sep
Institutional · hedge funds	81% · 128 funds	\$25.6bn at Q2; up from 114 funds
Notable exits, Q2-26	Duquesne, Appaloosa	the two most experienced macro holders
MSCI inclusion	31 Aug	done; visible in the 31 Aug volume
S&P 100 inclusion	21 Sep	announced 4 Sep (+12% that day); last mechanical bid
Company buyback	\$4.5bn FQ4 · \$15.5bn left	at ~24× trailing; a cycle-top signal (Micron 2018, 2022)
Analyst ratings, 3 months ago → now	3/15/3/1/0 → 4/16/3/0/1	Strong Buy / Buy / Hold / Sell / Strong Sell

SanDisk's obligations to Flash Ventures are unconditional; its customers' obligations to SanDisk are collateralised mostly off its balance sheet and untested

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Flash Ventures mechanics (FY26 10-K)	
Term	Disclosure
Ownership	49.9% SanDisk / 50.1% Kioxia; three JV entities (Yokkaichi Fabs 2-7, Kitakami K1/K2)
Wafer purchase	SanDisk buys ~50% of JV output at cost; pays 50% of fixed costs regardless of output
Commitments FY27-31	\$6.6bn purchase and funding commitments
Lease guarantees	\$923m of JV equipment leases guaranteed by SanDisk
Payments to Kioxia	\$1.2bn in FY26 for wafers, R&D and fab services
Own capex	~\$0.2bn/yr on SanDisk's books; the capital intensity is in the JV, off the SanDisk P&L until wafers are purchased
Japan plan (27 Aug 2026)	¥5tn (~\$31bn) with Kioxia; Kitakami Fab 3 output FY2029; SanDisk's share funded from peak cash flow

In a downturn SanDisk's cash cost is fixed, its wafer intake is contracted, and its selling price is the only variable. That is why the record on A2 has ten loss-making years in thirty-three.

Year	What happened to SanDisk / WDC Flash
2008	Operating margin -59%; JV output cut, fixed costs still paid; \$1.5bn+ impairments
2019	WDC Flash margin turned negative; Kioxia (Toshiba Memory) lost ¥170bn; JV under-utilisation charges
2023	Flash operating margin -22% ex-impairment; \$1.8bn impairment at the spin; net loss \$2.1bn

The NBM balances in the 10-K	
Balance (30 Jun 2026)	\$bn
Contract liabilities (customer prepayments)	1.24
Refundable customer deposits	1.50
Cash-backed guarantees on balance sheet	2.74
Headline "financial guarantees" (call)	16.5
Off-balance-sheet portion (third-party collateral)	13.8
Inventory · days	2.70 · ~178
Cash · debt	4.76 · 0.18

Inventory days are already rising
~178 days at FY26 year-end vs ~135 a year earlier, while the company reports allocation. Rising inventory in a "sold out" market is the classic pre-rollover tell (Micron 2018: 143 → 165 days into the peak). We compute hyperscaler and SanDisk inventory days from every 10-Q on the event path.

Open item: reconcile the ">50% FY27 / ~2/3 FY28" bit coverage (5 Aug call) with Citi's "50% by end-FY28" (8 Sep) against the replay.

Three years of cash burn, then \$11.5bn

Fiscal Year	Free cash flow, \$bn
FY23	-\$0.9bn
FY24	-\$0.5bn
FY25	-\$0.1bn
FY26	\$11.5bn

Every 2026 memory long-term agreement we can find		
Supplier	Structure	Tested?
SanDisk	10 NBMs, \$93.9bn floors, \$16.5bn guarantees, >4 yrs	No
Micron	\$100bn take-or-pay, \$22bn customer prepayments	No
Samsung	60-70% of capacity under 3-year-plus deals	No
SK hynix	HBM sold out 2026-27; NAND LTAs with hyperscalers	No
Kioxia	Multi-year eSSD supply agreements; volumes, not floors, disclosed	No
Apple (buyer)	Multi-year NAND supply from Kioxia/YMTC reported	-

What we say, and what we do not

- We say **"untested"**, not "always renegotiated." We found no primary documentation of 2019 or 2023 LTA renegotiations and do not claim any.
- Goldman on prior-generation agreements: "minimal enforceability – buyers could reduce volumes or breach with limited consequences."
- Two NBMs have already been re-opened in SanDisk's favour. Bilateral re-opening is evidence the terms are negotiable, in both directions.
- Our base case keeps the floors whole. Only the 20%-weighted bust case and 20% of Monte Carlo paths assume partial renegotiation.

What a renegotiation would look like, mechanically

Step	Consequence for SanDisk
Customer defers volume under a shortfall clause	Revenue timing slips; JV fixed cost unchanged
Bilateral re-price to "market plus" in exchange for extension	Floor economics fall; duration rises; the terminal is contracted <i>lower</i>
Draw on collateral	One-off cash; relationship and future volume at risk
Litigation	Multi-year; no precedent in memory of a supplier enforcing a floor against a hyperscaler